

Thesis Proposal: Functional Forecasting of Turkish Electricity Load Curves

Zeynep Bozoklu
GitHub repository

Motivation

This project studies day-ahead forecasting of Turkish electricity demand, measured hourly in MWh. The raw quantity of interest is not a single daily number, but the full 24-hour load profile: how demand evolves overnight, rises in the morning, peaks during the day, and falls again in the evening. Treating each day as a curve preserves this intraday shape.

The practical question is whether a functional time-series model can improve on the available official day-ahead forecast. The report therefore compares each proposed model with two reference points: a simple weekly benchmark, Y_{d-7} , and the official forecast, $\hat{Y}_d^{off}$. The weekly benchmark checks whether the model beats a very transparent rule based on the strong weekly rhythm of electricity use. The official forecast is the main benchmark because it represents the forecast that would be available in practice.

The proposed baseline contribution is a parsimonious functional forecasting model: daily curves are represented by FPCA scores, the scores are forecast with lagged functional dynamics and predetermined calendar, holiday, load-shape, and derivative features, and forecasts are evaluated in a rolling-origin design. A second-stage residual adjustment is reported separately as a possible extension to discuss rather than as the main specification.

Object and functional representation

Let $Y_d(h)$ denote Turkish electricity load on day d , hour $h = 0, \dots, 23$. The empirical object is the daily load curve

$$Y_d = \{Y_d(h) : h = 0, \dots, 23\}.$$

The cleaned data are stored as daily curve matrices

$$Y = [Y_d(h)]_{d,h}, \quad \hat{Y}^{off} = [\hat{Y}_d^{off}(h)]_{d,h}, \quad E = [Y_d(h) - \hat{Y}_d^{off}(h)]_{d,h},$$

with 24 hourly observations per retained day.

Each curve is represented by a B-spline basis,

$$Y_d(h) \simeq \sum_{m=1}^M c_{dm} B_m(h), \quad M = 12.$$

FPCA gives

$$Y_d(h) = \mu(h) + \sum_{k=1}^K \xi_{dk} \phi_k(h) + e_d(h), \quad K = 2.$$

The choice $K = 2$ is based on the empirical FPCA variance decomposition of the load curves. The first two load components explain 97.29% of total functional variation, exceeding the usual 95% cumulative-variance rule while keeping the score vector parsimonious. The first component mainly captures the overall daily level and the second adds the dominant intraday shape contrast. Components three and four explain only 1.11% and 0.99% of additional load variation, respectively, so they are not retained in the forecasting model. The official forecast-error FPCA is reported only as a diagnostic description of error shape; it is not used to choose K .

FPC	Load var.	Load cum.	Error var.	Error cum.
1	93.40%	93.40%	76.60%	76.60%
2	3.89%	97.29%	12.00%	88.60%
3	1.11%	98.40%	4.62%	93.22%
4	0.99%	99.39%	2.88%	96.09%

Let

$$\xi_d = (\xi_{d1}, \dots, \xi_{dK})', \quad \Phi(h) = (\phi_1(h), \dots, \phi_K(h))'.$$

Then

$$Y_d(h) = \mu(h) + \Phi(h)' \xi_d + e_d(h), \quad \hat{Y}_d(h) = \hat{\mu}(h) + \hat{\Phi}(h)' \hat{\xi}_d.$$

All forecasts are rolling-origin forecasts: for target day d , estimation uses only information dated $< d$. The evaluation period starts on 2023-01-01 and contains 26,304 hourly observations.

Proposed modeling strategy

The model sequence is intentionally incremental. Each specification asks whether one additional source of information explains a remaining part of the forecast error.

First, the daily curve is reduced to a small number of FPCA scores. This is useful because the 24 hourly observations are highly dependent: morning, afternoon, and evening demand move together in recurring ways. FPCA summarizes the main level and shape movements of the whole curve, so the forecasting problem becomes forecasting a few daily scores rather than 24 separate hourly loads.

The first FPCA models use lagged scores. A weekly lag is natural because electricity load has a strong day-of-week structure: Mondays tend to resemble previous Mondays, weekends resemble previous weekends, and so on. Adding a one-day lag allows the model to react to recent changes in demand level or shape that are not fully captured by the same weekday one week earlier. This motivates the FPCA VAR(1,7) score model.

The next additions are calendar, holiday, and load-shape predictors. These variables are included because Turkish electricity demand changes around weekends, national holidays, religious holidays, bridge days, and seasonal school or summer periods. Lagged load-shape variables, such as the previous day's mean, peak, range, and morning ramp, are included because they describe the current demand regime before the target day is forecast.

Derivative features are then added to describe how quickly the load curve is changing within the day. Two days can have similar average demand but different ramping behavior. The derivative summaries and targeted morning, evening, and end-of-day ramps are meant to capture these slope and transition patterns.

Forecasting specifications

Benchmarks:

$$\widehat{Y}_d^{SN}(h) = Y_{d-7}(h), \quad \widehat{Y}_d^{OFF}(h).$$

Separate weekly score autoregression:

$$\xi_{dk} = \alpha_k + \beta_k \xi_{d-7,k} + u_{dk}, \quad k = 1, \dots, K.$$

Weekly FPCA VAR:

$$\xi_d = a + A_7 \xi_{d-7} + u_d, \quad A_7 \in \mathcal{R}^{K \times K}.$$

FPCA VAR(1,7):

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + u_d, \quad A_1, A_7 \in \mathcal{R}^{K \times K}.$$

Augmented score model:

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + B z_d + u_d,$$

where z_d is predetermined at date d .

Residual-extension forecast:

$$\widehat{Y}_d^{res}(h) = \widehat{\mu}(h) + \widehat{\Phi}(h)' \widehat{\xi}_d + \lambda_d \widehat{r}_d(h).$$

Predictor construction

The largest predictor vector is

$$z_d = \left(z_d^{cal'}, z_d^{hol'}, z_d^{shape'}, z_d^{deriv'} \right)'.$$

Ramp and early-load variables:

$$R_{d-1}^m = Y_{d-1}(9) - Y_{d-1}(6), \quad R_{d-7}^m = Y_{d-7}(9) - Y_{d-7}(6),$$

$$L_{d-1}^e = \frac{1}{3} \sum_{h \in \{5,6,7\}} Y_{d-1}(h).$$

Calendar controls:

$$z_d^{cal} = \left(weekend_d, \sin \frac{2\pi doy_d}{365.25}, \cos \frac{2\pi doy_d}{365.25} \right)'.$$

Holiday controls include fixed holidays, religious holidays, holiday eve, post-holiday, near-holiday, and bridge-day indicators:

$$Bridge_d = 1\{dow_d \in \{Mon, Fri\}\} 1\{d \notin \mathcal{H}\} 1\{d-1 \in \mathcal{H} \text{ or } d+1 \in \mathcal{H}\}.$$

Lagged load-shape variables:

$$\bar{Y}_{d-1} = \frac{1}{24} \sum_{h=0}^{23} Y_{d-1}(h), \quad P_{d-1} = \max_h Y_{d-1}(h), \quad M_{d-1} = \min_h Y_{d-1}(h),$$

$$G_{d-1} = P_{d-1} - M_{d-1}, \quad \bar{Y}_d^{(q)} = \frac{1}{q} \sum_{j=1}^q \left(\frac{1}{24} \sum_{h=0}^{23} Y_{d-j}(h) \right), \quad q \in \{7, 28\}.$$

Derivative features are computed from the smoothed curve,

$$DY_d(h) = \frac{d}{dh} Y_d(h),$$

and summarized by

$$\overline{DY}_{d-1}, \quad SD(DY_{d-1}), \quad \max_h DY_{d-1}(h), \quad \min_h DY_{d-1}(h), \quad \max_h DY_{d-1}(h) - \min_h DY_{d-1}(h).$$

Targeted intraday changes:

$$R_{d-1}^e = Y_{d-1}(7) - Y_{d-1}(5), \quad R_{d-1}^{ev} = Y_{d-1}(20) - Y_{d-1}(17), \quad R_{d-1}^{end} = Y_{d-1}(23) - Y_{d-1}(20).$$

Evaluation

For model m ,

$$e_{dh}^{(m)} = Y_d(h) - \hat{Y}_d^{(m)}(h).$$

$$MAE_m = \frac{1}{N} \sum_{d,h} |e_{dh}^{(m)}|, \quad RMSE_m = \left[\frac{1}{N} \sum_{d,h} (e_{dh}^{(m)})^2 \right]^{1/2}, \quad Bias_m = \frac{1}{N} \sum_{d,h} e_{dh}^{(m)}.$$

$$Gain_m^{MAE} = 100 \frac{MAE_{OFF} - MAE_m}{MAE_{OFF}}, \quad Gain_m^{RMSE} = 100 \frac{RMSE_{OFF} - RMSE_m}{RMSE_{OFF}}.$$

Preliminary results

The rows below should be read as a model-building path. The first two rows are benchmarks. Each later row adds one modeling idea: FPCA score dynamics, calendar and ramp information, holiday/load-shape information, and derivative information.

Model	MAE	RMSE	Bias	MAE gain	RMSE gain
Seasonal naive Y_{d-7}	1912.46	3129.98	39.85	-57.60%	-79.85%
Official forecast	1213.50	1740.30	466.87	0.00%	0.00%
FPCA VAR(1,7) scores	1442.61	2018.30	160.13	-18.88%	-15.97%
FPCA VAR(1,7) + ramp/calendar	1238.37	1716.30	110.12	-2.05%	1.38%
FPCA VAR(1,7) + holiday/load-shape	1000.62	1353.56	-54.71	17.54%	22.22%
FPCA VAR(1,7) + holiday/load-shape/derivative	980.38	1324.40	102.56	19.21%	23.90%

The final model improves on the official forecast by 19.21% in MAE and 23.90% in RMSE. Relative to the weekly seasonal naive benchmark,

$$100 \frac{1912.46 - 980.38}{1912.46} = 48.74\%.$$

Discussion: K and residual correction

The main report uses $K = 2$, because this follows the 95% cumulative-variance rule for the load FPCA and gives the most parsimonious score model. The cost of this choice is some loss of accuracy relative to $K = 4$. The $K = 4$ no-residual model has lower overall MAE and RMSE, and it beats the official forecast in 23 of 24 hours by MAE and 24 of 24 hours by RMSE. The $K = 2$ no-residual model wins in 19 of 24 hours by MAE and 21 of 24 hours by RMSE.

Specification	K	Resid. step	MAE	RMSE	MAE gain	RMSE gain
Holiday/load/derivative	2	No	980.38	1324.40	19.21%	23.90%
Holiday/load/derivative	2	Yes	652.90	982.60	46.20%	43.54%
Holiday/load/derivative	4	No	800.99	1109.69	33.99%	36.24%
Holiday/load/derivative	4	Yes	640.10	967.82	47.25%	44.39%

The table suggests a tradeoff between parsimony and predictive accuracy. The $K = 2$ specification is the conservative baseline, while $K = 4$ retains additional intraday shape variation and improves the no-residual forecast. I therefore keep $K = 2$ as the main specification, but report $K = 4$ as a robustness check.

The second-stage residual correction should be interpreted even more cautiously, especially its methodological appropriateness. The closest related papers are not functional forecasting papers, but papers on residual post-processing and error compensation in electricity forecasting. Xie and Hong (2016) use forecast combination and residual simulation in probabilistic electricity load forecasting. Wang et al. (2012) use residual modification models to improve seasonal ARIMA electricity demand forecasts. Kontogiannis et al. (2022) estimate hourly residual error sequences to refine day-ahead electricity price forecasts. Ghimire et al. (2024) also use a two-step forecasting design in which an initial electricity price forecast is improved by an error-compensation model. These papers motivate the idea of

learning from past forecast errors, but I do not yet claim that they validate the exact functional residual correction used here. For this reason, I treat the residual correction as a promising empirical extension rather than as the main model.

For target day d , the base model is first estimated exactly as before. Then fitted residual curves are computed only for days strictly before the target day,

$$\hat{e}_s(h) = Y_s(h) - \hat{Y}_s^{base}(h), \quad s < d.$$

The residual second step estimates a correction curve $\hat{r}_d(h)$ from these past residuals and applies

$$\hat{Y}_d^{res}(h) = \hat{Y}_d^{base}(h) + \lambda_d \hat{r}_d(h).$$

Both the residual regression and the validation choice of λ_d use only information dated strictly before d . Thus the extension is still rolling-origin and is designed to avoid information leakage.

Model	MAE	RMSE	Bias	MAE gain	RMSE gain
Best base model	980.38	1324.40	102.56	19.21%	23.90%
Base model + residual second step	652.90	982.60	90.03	46.20%	43.54%

Conclusion

This preliminary report treats Turkish hourly electricity load as daily curves and evaluates whether an FPCA-based rolling-origin model can improve on the official day-ahead forecast. The conservative $K = 2$ specification improves on the official forecast by 19.21% in MAE and 23.90% in RMSE.

The results suggest that intraday load-shape information, calendar effects, holidays, and derivative/ramp features contain useful predictive information. However, the model specification is still preliminary. Further work is needed to decide how many components should be retained, whether the residual correction is methodologically appropriate, and how the forecast gains can be connected to the economic intuition behind intraday demand patterns and system operation.

References

- Xie, J. and Hong, T. (2016). GEFCom2014 probabilistic electric load forecasting: An integrated solution with forecast combination and residual simulation. *International Journal of Forecasting*, 32(3), 1012–1016. doi:10.1016/j.ijforecast.2015.11.005.
- Wang, Y., Wang, J., Zhao, G. and Dong, Y. (2012). Application of residual modification approach in seasonal ARIMA for electricity demand forecasting: A case study of China. *Energy Policy*, 48, 284–294. doi:10.1016/j.enpol.2012.05.026.
- Kontogiannis, D., Bargiotas, D., Daskalopulu, A., Arvanitidis, A. I. and Tsoukalas, L. H. (2022). Error compensation enhanced day-ahead electricity price forecasting. *Energies*, 15(4), 1466. doi:10.3390/en15041466.
- Ghimire, S., Deo, R. C., Casillas-Pérez, D. and Salcedo-Sanz, S. (2024). Two-step deep learning framework with error compensation technique for short-term, half-hourly electricity price forecasting. *Applied Energy*, 353, 122059. doi:10.1016/j.apenergy.2023.122059.
- Kokoszka, P. and Reimherr, M. (2017). *Introduction to Functional Data Analysis*. CRC Press.